

LEAF-EB

Long-Term Energy Analysis Framework — Electricity Balance

USER MANUAL



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About this manual

This manual is intended to be used while building a LEAF-EB study. It explains not only where a field is written in YAML, but what that field means in the calculation, which alternatives are available, which combinations are valid and which settings should remain explicit for reproducibility.

LEAF-EB constructs and evaluates long-term electricity scenarios from user-defined assumptions and historical data. Those assumptions can be entered directly in LEAF-EB or adapted from planning studies, optimization models, policy targets or published scenarios. LEAF-EB does not require the scenario to exist before the framework is used; it also does not optimize the future generation portfolio.

Recommended use. For a first run, execute Examples 01–06 in order. For a new study, use Sections 3–11 to construct the YAML. Appendix A is the field-by-field reference and Appendix B lists accepted units.

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1 What LEAF-EB does

1.1 Purpose

LEAF-EB links long-term assumptions with daily or hourly electricity-system operation. It constructs future demand and generation series, places them on a chronological time axis using historical temporal information, then evaluates the electricity balance sequentially. Storage state, commodity inventories and nuclear operating/refueling state can therefore carry information from one period to the next.

1.2 What is defined by the user

- Study dates and output location.
- Historical data and its time resolution.
- Demand target and dated demand factors.
- Generation sources, shares or explicit capacity additions, and dated changes.
- Hourly operating rules for sources that are not purely profile-following.
- Storage, interconnections, commodity pathways and dispatch order when used.
- Nuclear deployment, operating rule and refueling basis when used.
- Which time series are stochastic and how Monte Carlo simulations are executed.
- Output detail and whether emissions are calculated.

1.3 What LEAF-EB does not decide

- It does not optimize the generation portfolio or choose an economically optimal expansion plan.
- It does not solve network-constrained unit commitment, market clearing or AC/DC power flow.
- It does not infer missing physical parameters for BESS, nuclear operation or commodity pathways when those parameters affect the calculation.
- An additional generation requirement means represented generation is insufficient to meet demand at that time step; it is not by itself a probabilistic reliability index.

2 Installation and first run

2.1 Install

```
python -m pip install -U pip
python -m pip install -r requirements.txt
```

LEAF-EB requires CPython 3.11 or newer. Run the commands from the project root, the directory that contains Runner.py. Relative paths in the YAML are resolved from that project root.

For development and the complete test suite:

```
python -m pip install -r requirements-dev.txt
```

2.2 List and run inputs

```
python Runner.py --list
python Runner.py Example_01_basic
python Runner.py Example_02_monte_carlo
```

The .yml extension is optional. Running Runner.py with no argument opens the scenario selector.

2.3 Recommended first sequence

1. Run Example_01_basic without changing anything.
2. Run Example_02_monte_carlo to verify stochastic execution.
3. Run Example_03_bess and Example_04_commodities to understand the dispatch sequence.
4. Run Examples 05 and 06 together to see how nuclear operating strategy changes refueling timing.
5. Only after those examples work, copy the closest YAML and replace one layer at a time.

3 How to build an input correctly

3.1 Do not begin with every option enabled

The safest workflow is progressive. A LEAF-EB input is easier to diagnose when the deterministic electricity balance is valid before storage, commodity conversion, nuclear operating constraints or Monte Carlo uncertainty are added.

1. Prepare and validate the historical dataset.
2. Define scenario dates, temporal resolutions and demand.
3. Add generation sources and run a deterministic case.
4. Assign hourly operating modes only to sources that require them.
5. Add BESS/interconnections/commodities and verify the dispatch order.
6. Add nuclear deployment/refueling if required.
7. Enable Monte Carlo last, first with a small sample and then with the study sample size.

3.2 Required top-level structure

```
scenario:
  ...
historical_data:
  ...
projection:
  ...
simulation:
  ...
demand:
  ...
sources:
  ...
commodities_input: # optional
  ...
output:           # optional
  ...
```

Block	Required?	Purpose
scenario	Yes	Study dates and output folders.
historical_data	Yes	Historical demand/generation file, resolution and common unit.
projection	Yes	Resolution used to construct future demand/generation and temporal variability settings.
simulation	Yes	Operational resolution, capacity tolerance, initial states and Monte Carlo settings.
demand	Yes	Demand target and dated scaling factors.
sources	Yes	Generation-source definitions and operating rules.
commodities_input	No	BESS, interconnections, commodities and sequential dispatch.
output	No	Output level, event threshold and emissions switch.

Canonical schema. Use the field names documented here. LEAF-EB rejects unknown source, commodity, BESS and output fields rather than silently guessing what they mean.

Execution note. The sequential electricity-balance calculation is run only when `commodities_input` is present with `run_commodities: true`. For a balance-only study, the block can contain only that switch; BESS and commodity production are not required.

3.3 YAML comments

Everything after # on a YAML line is a comment and is ignored by LEAF-EB.

```
capacity_tolerance: "0 %" # No allowed capacity exceedance.
seed: 12345              # Reproducible Monte Carlo seed.
```

4 Historical data and temporal resolutions

4.1 Required historical-file structure

```
Date,Demand,Solar,Wind,Hydro
2020-01-01 00:00,1000,250,300,180
2020-01-01 01:00,970,0,320,175
...
```

- CSV and XLSX are supported.
- One date/time column is required; its name is set by `historical_data.date_column`.
- Demand is required.
- Every non-custom source in sources must have a matching historical column.
- Demand and configured historical generation values must be numeric and non-negative.
- Timestamps must be unique and form a complete regular daily or hourly time axis.
- All energy columns in the historical file use one common unit declared by `historical_data.unit`.

4.2 The three resolutions

Resolution	Where set	Meaning
Historical processing	<code>historical_data.processing_resolution</code>	How the observed record is processed for pattern/residual construction.
Projection	<code>projection.resolution</code>	Time step used to construct the future demand and generation scenario series.
Simulation	<code>simulation.resolution</code>	Time step used for sequential electricity balance and operational rules.

4.3 Common supported combinations

Historical data	Processing	Projection	Simulation	Extra requirement
Daily	daily	daily	daily	None.
Hourly	hourly	daily	hourly	Hourly profiles are built from the historical file.
Hourly	hourly	hourly	hourly	No daily-to-hourly reconstruction is needed.
Hourly	daily	daily	daily	Hourly record can be aggregated for daily processing.
Daily	daily	daily	hourly	<code>simulation.hourly_profile_file</code> with exactly 8760 rows.

Rule. Hourly projection requires hourly historical processing. A daily simulation cannot consume an hourly projection. `projection.variability.resolution`, when explicitly supplied, must match `projection.resolution`.

4.4 External hourly profile

Use this only when the historical file is daily but the operational simulation must be hourly.

```
simulation:
  resolution: hourly
  hourly_profile_file: Inputs/data/my_8760_profile.xlsx
```

The profile must contain exactly 8760 rows and a Demand column. Add columns for every standard source that must be expanded to hourly operation.

5 Scenario, projection and simulation blocks

5.1 scenario

```
scenario:
  folder: Outputs
  subfolder: MyScenario
  start_date: '2030-01-01'
  end_date: '2050-12-31'
```

Field	Required	Options / rule	Meaning
folder	Yes	Non-empty text	Parent results folder.
subfolder	Yes	Non-empty text	Scenario-specific result folder.
start_date	Yes	Valid date/time	First simulated date.
end_date	Yes	Valid date/time; >= start_date	Last simulated date.

5.2 projection

```
projection:
  resolution: daily
  variability:
    enabled: true
    resolution: daily
    mode: M
  plot_residuals: false
  save_plots: false
```

Field	Default / options	Use
resolution	daily hourly	Future scenario-series time step.
save_plots	false	Save projection diagnostic figures.
pattern_file	Pattern.xlsx	Name of the generated/used pattern workbook; normally leave default.
variability.enabled	false	Build residual history; required when simulations > 0.
variability.resolution	projection.resolution	Must match projection resolution.
variability.mode	global; use M/monthly for monthly fallback parameters	Controls whether Gaussian fallback parameter files are global or monthly. Stationary-bootstrap history remains empirical.
variability.plot_residuals	false	Save residual diagnostic plots.

5.3 simulation

```
simulation:
  resolution: hourly
  capacity_tolerance: "0 %"
  console_output: standard
  initial_conditions:
    bess_state_of_charge: 0.0
    commodity_inventory:
      CH4: 0.0
```

Field	Default / options	Meaning
resolution	daily hourly	Operational balance time step.
capacity_tolerance	0 %	Global percentage by which inferred/installed source capacity may be exceeded before clipping; source-specific value overrides it.
console_output	quiet standard detailed	Amount of runtime information printed.
hourly_profile_file	none	Required for daily historical data -> hourly simulation.
initial_conditions.bess_state_of_charge	0.0; 0-1	Initial battery state as a fraction of energy capacity.
initial_conditions.commodity_inventory	empty mapping	Initial stored amount for each configured commodity; values must be >= 0.

6 Demand

6.1 Meaning of target_production and values

`demand.target_production` defines the reference demand level. It is an energy-per-period quantity, for example "6000 MWh/day" or "200 TWh/year". LEAF-EB converts it to the configured projection time step. `demand.values` are dimensionless dated factors applied to that reference level.

```
demand:
  target_production: "200 TWh/year"
  model: linear
  balance: 1.0
  values:
    '2030-01-01': 0.75
    '2040-01-01': 0.90
    '2050-01-01': 1.00
```

Interpretation. At a dated factor of 1.00 the demand target is the full `target_production` level. A factor of 0.75 represents 75% of that target. Dated values are interpolated according to model.

Field	Required	Options / constraints	Meaning
target_production	Yes	Energy per hour/day/year; > 0	Reference demand level.
model	Yes	linear exponential	Interpolation between dated factors.
values	Yes	At least two dated non-negative factors	Demand scaling factors.
balance	No	> 0; default 1.0	Final-year mean production/demand target applied as a common scale to non-custom generation before custom replacement. Keep 1.0 unless the study intentionally represents systematic over- or under-supply.
anchor	No	auto mean median last max	How the future demand path is connected to recent historical demand.

6.2 Linear versus exponential interpolation

Model	Use when	Behavior
linear	Default; smooth absolute change between milestones.	Straight interpolation between dated factors; segments touching zero remain well defined.
exponential	Growth or decline should be proportional between strictly positive milestones.	Interpolates by a constant ratio; if either endpoint is zero, LEAF-EB falls back to linear for that segment.

Milestone boundaries. If the first dated value is later than `scenario.start_date`, LEAF-EB bridges from the historical anchor to that first milestone. After the last milestone, the last factor is held constant through the end of the scenario.

7 Generation sources

7.1 Standard source: share-based projection

```
sources:
  Wind:
    share: 0.30
    capacity_factor: 0.38
    model: linear
    values:
      '2030-01-01': 0.80
      '2050-01-01': 1.00
```

For a standard source, share is an energy share relative to the demand target path, not a share of installed power. LEAF-EB first forms a source reference from demand × share. The dated values then scale that source reference and historical patterns distribute it in time. capacity_factor is used to infer installed capacity and enforce physical generation limits.

Special case — share: 0. A zero-share standard source is treated as a closure path: LEAF-EB connects the positive historical anchor to zero at the first date listed in values and keeps the source at zero thereafter. The numerical milestone factor does not set the closure magnitude in this special case.

Example. If demand at the target date is 100 TWh/year, share = 0.30 and values = 1.0, the source reference is 30 TWh/year before temporal patterning and final balance scaling.

7.2 Core source fields

Field	When used	Options / rule	Meaning
share	Standard sources; optional when capacity_additions is used	>= 0	Energy share of demand reference.
capacity_factor	Required unless capacity_additions is used	(0, 1]	Converts energy projection to inferred installed capacity.
capacity_tolerance	Optional	0–100 %	Source-specific capacity exceedance allowance; overrides simulation.capacity_tolerance.
model	Always	linear exponential custom	Projection construction method.
values	linear/exponential	At least two dated non-negative factors	Milestone factors.
anchor	Optional	auto mean median last max	Historical starting-level rule.
limit	Optional	>= 0	Dynamic production ceiling for specialized source schedules.
emission_factor_co2	Required if output.emissions = true	e.g. 12 gCO ₂ e/kWh	Source lifecycle/direct factor used by the configured emissions calculation.
hourly_operation	Optional	profile_preserving dispatchable must_run load_following	Operational treatment in an hourly simulation.
dispatch_priority	Conditional	Positive integer; no duplicates within operation class	Priority for sources whose hourly output can be adjusted.

7.3 Anchoring to historical data

```
anchor:
  method: auto
  # window is not needed for auto
```

Method	Meaning
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Method	Meaning
auto	Uses the mean over the latest annual historical cycle as the starting level. After temporal patterning, LEAF-EB checks the forecast boundary against empirical historical transitions and, when needed, applies a smooth data-driven bridge while preserving calendar-year mean energy. No manual window is required.
mean	Mean of the last window observations.
median	Median of the last window observations.
last	Last available observation.
max	Maximum of the last window observations.

Automatic anchoring is reproducible. Resolved_Config.yaml records the effective anchor value and any automatic boundary bridge selected by the calculation. Manual methods (mean, median, last and max) remain explicit user choices.

For mean, median and max, window is a positive integer number of historical observations. Use a manual anchor only when there is a documented reason; Resolved_Config.yaml records the effective anchor information.

7.4 Custom sources: add versus replace

model: custom is used when the future source is controlled by dated changes rather than a share-based interpolation.

custom_mode	Meaning	Typical use
add	The custom source is added to the scenario; other source projections are not reduced automatically.	New capacity added on top of an existing generation scenario, or explicit retirements/additions of an existing fleet.
replace	The custom source is added and configured source groups are reduced annually so that its energy replaces theirs while preserving their within-year profiles.	Nuclear replacing wind/solar energy in a controlled comparison.

7.5 Custom source with explicit capacity additions

```
Nuclear:
  model: custom
  custom_mode: add
  initial_capacity: "7000 MW"
  capacity_additions:
    '2035-12-31': "-1000 MW"
    '2040-12-31': "-2000 MW"
  reference_generation:
    capacity_factor: 0.83
```

- capacity_additions is a date -> power change mapping. Positive values add capacity; negative values retire it.
- The cumulative installed capacity may never become negative.
- capacity_additions requires model: custom.
- When refueling is not configured, reference_generation.capacity_factor is required to construct reference generation from installed capacity.
- When refueling is configured, reference_generation must be omitted because availability is derived from the refueling model.

7.6 Custom source using custom_data

```
Firm_Source:
  share: 0.0
  capacity_factor: 0.90
  model: custom
  custom_mode: add
  custom_data:
```

```
'2035-01-01': "500 MWh"  
'2040-01-01': "250 MWh"
```

`custom_data` applies dated additive changes to the custom schedule. It cannot be used together with `capacity_additions`. Use explicit units when the entries are dimensional energy values.

7.7 Replacement groups

```
custom_mode: replace  
replaces:  
- [Wind, Solar]  
- [Gas]
```

Replacement groups are evaluated in order. Within a group, annual replacement is proportional to the annual generation available from the listed sources. A source may appear only once, and the custom source cannot replace itself.

Reference versus realized operation. The replacement amount is calculated from the custom source reference series during scenario construction, before later hourly dispatch or load-following. A load-following source can therefore realize different generation without retroactively changing the annual wind/solar reduction. This is useful when structural replacement must remain common across operating-strategy comparisons.

Insufficient replacement energy. If the ordered replacement groups do not contain enough annual generation to absorb the requested replacement, LEAF-EB leaves the unmatched part as unallocated replacement and reports it in diagnostics rather than making any source negative.

8 Hourly operation and dispatch

8.1 Choosing hourly_operation

Option	What LEAF-EB does	Requires
profile_preserving	Retains the reconstructed/sampled hourly shape of that source instead of treating it as freely dispatchable.	No dedicated sub-block; dispatch_priority optional.
dispatchable	Uses available production only when the residual electricity requirement calls for it.	dispatch_priority is required.
must_run	Operates at a fixed fraction of available power while online; outages/refueling can reduce availability.	must_run.power_fraction.
load_following	Adjusts output within configured power limits according to its energy policy/control mode.	load_following block; dispatch_priority optional; refueling rules may impose additional constraints.

Do not assign an hourly mode unnecessarily. Wind, solar and other weather-following series can normally use their reconstructed profile without an explicit operating block. Use hourly_operation only when the technology requires a specific operational rule.

8.2 Dispatchable sources

```
Gas:
share: 0.10
capacity_factor: 0.50
model: linear
hourly_operation: dispatchable
dispatch_priority: 1
values:
'2030-01-01': 1.0
'2050-01-01': 1.0
```

dispatch_priority is a positive integer and may not be repeated among dispatchable sources. Lower-priority-number sources are used earlier in their operation class.

8.3 Must-run sources

```
hourly_operation: must_run
must_run:
power_fraction: 1.0
```

power_fraction must be > 0 and <= 1. A value of 1.0 means full available power while online. Availability can still fall because of a refueling outage.

8.4 Load-following

```
hourly_operation: load_following
dispatch_priority: 1
load_following:
energy_policy: follow_residual
control_mode: constrained_hourly
minimum_power_fraction: 0.50
maximum_power_fraction: 1.00
ramp_up_rate: "0.50 /h"
ramp_down_rate: "0.25 /h"
deep_reduction_threshold_fraction: 0.80
deep_reduction_request_duration: "2 h"
max_deep_reduction_cycles: "2 /day"
```

Field	Options / constraint	Meaning
-------	----------------------	---------

Field	Options / constraint	Meaning
energy_policy	preserve_annual follow_residual	Whether operation attempts to preserve annual energy or follows residual requirement causally.
control_mode	direct constrained_hourly	direct applies power bounds; constrained_hourly also applies ramps/deep-reduction rules.
minimum_power_fraction	0–1	Minimum output fraction when operating.
maximum_power_fraction	(0,1]	Maximum output fraction.
ramp_up_rate	(0,1] per hour	Maximum upward fractional change per hour for constrained_hourly.
ramp_down_rate	(0,1] per hour	Maximum downward fractional change per hour.
deep_reduction_threshold_fraction	between min and max	Threshold used to identify a deep reduction.
deep_reduction_request_duration	positive whole hours	Persistence required before a deep reduction is accepted.
max_deep_reduction_cycles	non-negative integer per day	Daily limit on deep-reduction cycles.

Burnup-driven refueling. If load-following refueling is driven by target_burnup/EFPD, energy_policy must be follow_residual. preserve_annual is not compatible with that causal refueling formulation.

9 Nuclear deployment, operation and refueling

9.1 Recommended: use a technology template

```
Nuclear_SMR300:
  technology_template:
    file: data/Nuclear_Technologies.yaml
    name: smr_300_burnup60
  custom_mode: add
  hourly_operation: load_following
  dispatch_priority: 1
  capacity_additions:
    '2035-01-01': "600 MW"
```

The template supplies reactor-specific values and operating/refueling defaults. Explicit fields in the scenario source override template values because scenario-specific settings win during the merge.

Template	Refueling basis	Purpose
smr_300_reference	24 month calendar	Generic 300-MW reference template.
smr_300_calendar60	24.7986403728 month calendar	Calendar interval corresponding to about 60 GWd/tHM at full-power equilibrium for the reference data.
smr_300_burnup60	60 GWd/tHM	Burnup/EFPD-driven refueling; actual operation affects outage timing.

9.2 Refueling fields

Field	Required / options	Meaning
unit_capacity	Required when refueling is defined; >0	Electrical capacity of one unit. Installed fleet capacity is divided into units using this value.
refueling.mode	offline online; default offline	Whether refueling removes unit capacity from service. Calendar cycle is not used with online mode.
refueling.schedule	auto staggered; default auto	auto keeps each equivalent unit on its own refueling clock and allows overlapping outages. staggered shifts outages to enforce a non-overlapping fleet schedule when that schedule is feasible.
refueling.outage_duration	Required for offline; whole positive days	Duration of each outage.
refueling.fuel_batches	Required; integer >=1	Number of equal fuel batches represented in the core.
refueling.operating_cycle	Calendar basis only	Planned online operating interval from commissioning or the end of one outage to the start of the next outage. outage_duration is additional to this interval.
fuel_cycle.target_burnup	Burnup basis only	Full fuel-residence/equilibrium burnup target used to derive the EFPD cycle; mutually exclusive with operating_cycle. Startup discharges can be below this value.
fuel_cycle.thermal_power	Required for calendar/burnup fuel accounting	Thermal power of the modeled unit.
fuel_cycle.core_fuel_mass	Required with thermal_power	Heavy-metal mass in the modeled core.

Exactly one basis. A refueling source must define exactly one basis: refueling.operating_cycle OR fuel_cycle.target_burnup. Defining both, or neither, is rejected.

Fleet consistency. At every time step, installed fleet capacity must be an integer multiple of unit_capacity; otherwise LEAF-EB stops rather than creating fractional reactor units.

Fuel batches. fuel_batches represents equal-mass core batches. The simplified reload fraction is $1/\text{fuel_batches}$ and the discharged mass at each refueling is $\text{core_fuel_mass}/\text{fuel_batches}$. Newly commissioned cores retain startup-batch effects; LEAF-EB does not assign equilibrium burnup to the first discharge.

Burnup-driven cycle. LEAF-EB derives full residence EFPD as $\text{target_burnup} \times 1000 \times \text{core_fuel_mass} / \text{thermal_power}$, then divides that residence by `fuel_batches` to obtain the batch-refueling EFPD threshold. EFPD accumulates from actual generation, so reduced output can delay the next offline outage in calendar time.

Online refueling. mode: online is a specialized idealization: it is burnup/EFPD-based and tracks fuel exposure without removing electrical capacity for an outage. `refueling.operating_cycle` is therefore not used with online mode.

9.3 Fixed-calendar versus burnup-driven refueling

Basis	Outage timing depends on actual generation?	Best for
Calendar	No. The configured calendar interval determines the operating cycle.	Comparisons where outages follow a fixed time schedule.
Burnup / EFPD	Yes. Lower actual generation accumulates effective full-power exposure more slowly and can delay refueling.	Studies linking load-following operation to fuel use/refueling timing.

10 Monte Carlo uncertainty

10.1 Minimal stochastic block

```
projection:
  variability:
    enabled: true
    resolution: daily
    mode: M

simulation:
  monte_carlo:
    simulations: 200
    workers: 4
    seed: 12345
    confidence_level: 0.95
    preserve_annual_targets: false
    technology_uncertainty: false
    resume: true
    sources:
      - Demand
      - Wind
      - Solar
```

Field	Default / options	Meaning
simulations		0 Number of stochastic simulations; 0 means deterministic only.
workers	1; positive integer	Maximum worker count; memory safeguards may reduce effective parallelism.
seed	12345; non-negative integer	Random seed for reproducibility and paired scenario studies.
confidence_level	0.95; 0<value<1	Central interval reported by applicable summaries.
sources	Required if simulations>0	Explicit list chosen from Demand and configured generation sources.
preserve_annual_targets	false	If true, sampled annual energy is rescaled to the deterministic annual target; if false, annual stochastic energy may vary.
technology_uncertainty	false	If true, supported [min, mode, max] technology parameters in Database.yaml are sampled triangularly once per simulation. If false, their mode is used. This is separate from temporal uncertainty.
resume	false	Reuse valid completed simulations after interruption.
temporary_output_format	parquet; parquet pickle/pkl	Intermediate MC storage format.
keep_analysis_temp	false	Keep or remove temporary consolidation folders.
save_perturbed_forecasts.first_monte_carlo	true	Save the first stochastic forecast for inspection.
save_perturbed_forecasts.all_monte_carlo	false	Save every perturbed forecast; can become very large.

10.2 What Monte Carlo samples

- Simulation 0 is the deterministic reference. Monte Carlo simulations are numbered from 1 onward.
- `simulation.monte_carlo.sources` explicitly selects the demand/generation series that receive temporal perturbations. Selected historical residual series are sampled jointly, so contemporaneous relationships among them are retained.
- Temporal uncertainty uses an empirical stationary bootstrap with a data-driven common block length. At block restarts, daily histories match target month and weekday; direct-hourly histories match target month and hour.
- When the projection is daily but hourly historical data are available for an hourly simulation, the bootstrap-selected historical day also supplies the normalized 24-hour shape for the selected stochastic series.
- Custom sources are governed by their dated schedules/operating rules and dispatchable sources are allocated against the residual requirement; they should not normally be selected for historical-residual perturbation.
- Capacity limits remain binding after stochastic reconstruction. Energy clipped by a physical capacity limit is reported and is not shifted to a later time step.

Interannual variability. When at least two complete historical years are available, LEAF-EB also samples a linked annual variability regime from the historical record. With less history, this additional interannual modulation is not inferred.

Technology uncertainty. Database.yaml uses [value] for a fixed parameter and [min, mode, max] for a triangular distribution. Deterministic runs and temporal-only Monte Carlo use the fixed value or triangular mode. With technology_uncertainty: true, each supported triangular parameter is sampled once per Monte Carlo simulation and reused consistently within that simulation.

10.3 preserve_annual_targets is a methodological choice

This switch should not be selected only because one option produces narrower uncertainty bands. It changes what uncertainty represents:

- false: sampled historical conditions can modify both timing and annual energy relative to the deterministic case.
- true: stochastic timing is retained but each annual series is forced back to the deterministic annual total.

Report the choice explicitly in any analysis that uses Monte Carlo results.

10.4 Paired scenario comparisons

For paired comparisons, keep the same seed, stochastic source list and simulation IDs across scenarios. This makes differences less contaminated by different sampled historical sequences.

11 BESS, interconnections and commodities

11.1 BESS

Important switch. Despite its historical name, `commodities_input.run_commodities` controls execution of the sequential electricity-balance/flexibility calculation. If `commodities_input` is absent or `run_commodities` is false, `Runner.py` stops after building the scenario series; BESS and interconnections are not dispatched. Use `run_commodities: true` even for a balance-only or BESS-only study.

```
commodities_input:
  run_commodities: true
  BESS:
    Total:
      model: linear
      values:
        '2030-01-01': "5000 MW"
        '2050-01-01': "12000 MW"
      duration: "4 h"
      efficiency: 0.90
      lifetime:
        calendar_years: 15
        cycle_life_efc: 5000
```

Field	Required / options	Meaning
model	constant step linear	Installed BESS power schedule.
value	Required for constant	Constant installed power.
values	Required for step/linear	Dated installed-power values.
duration	Required; >0	Rated energy capacity = installed power × duration.
efficiency	Required; (0,1]	Delivery efficiency applied on discharge; charge is stored before this delivery loss in the current model.
lifetime.calendar_years	Optional; >0	Calendar-life diagnostic/replacement assumption.
lifetime.cycle_life_efc	Optional; >0	Equivalent-full-cycle life diagnostic/replacement assumption.

Named batteries. The recommended form is `BESS: {name: {...}}`. Multiple named assets are supported. `simulation.initial_conditions.bess_state_of_charge` is one global fraction applied to every named BESS asset at initialization. If `lifetime` is used for aggregate replacement diagnostics, all named assets must define compatible lifetime assumptions.

11.2 Dispatch order

```
Dispatch_Order:
  surplus:
    - BESS
    - Interconnections
    - Commodities_Production
  deficit:
    - BESS
    - Interconnections
    - Fuel_to_Electricity
```

Balance side	Allowed stages	Constraint
surplus	BESS, Interconnections, Commodities_Production	Commodities_Production must be last if present.
deficit	BESS, Interconnections, Fuel_to_Electricity	Fuel_to_Electricity must be last if present.

The YAML key `deficit` is retained in the configuration schema. In physical interpretation it means an additional generation requirement to meet demand.

11.3 Interconnections

```
Interconnections:
  EU:
```

```

mode: MW
import:
  model: linear
  values:
    '2030-01-01': "5000 MW"
    '2050-01-01': "8500 MW"
  availability: 0.90
export:
  model: constant
  value: "6000 MW"
  availability: 0.85

```

Field	Options	Meaning
mode	MW fraction	Absolute MW power or fraction of total installed generation capacity.
import/export.model	constant step linear	Directional capacity schedule.
value	constant model	Directional constant value.
values	step/linear	Dated directional values.
availability	0–1 scalar OR 12-month mapping	Fraction of directional capacity available at each time.

Both directions are required. Every named interconnection must explicitly define import, export, mode and availability. This prevents a missing direction from being interpreted as an unintended physical default.

11.4 Commodity production, storage and reconversion

```

database_path: data/Database.yaml
Commodities_Production:
  CH4:
    share: 1.0
    technology: M
    inputs:
      H2O: MED
      H2: AE
      CO2: DACG
  Commodity_Storage:
    CH4:
      max_age: "20 year"
      max_storage_quantity: 1000000000
  Fuel_to_Electricity:
    CH4:
      share: 1.0
      technology: CCGT

```

- Production shares across configured commodities must sum to 1.
- Production technology names and required upstream inputs must exist in data/Database.yaml.
- Every produced commodity needs an explicit storage block with max_age and storage capacity.
- Fuel_to_Electricity requires an explicit share and reconversion technology.
- Technology uncertainty is controlled separately by simulation.monte_carlo.technology_uncertainty.

Commodity quantity units. Commodity_Storage.<commodity>.max_storage_quantity and simulation.initial_conditions.commodity_inventory.<commodity> use the physical quantity unit declared for that commodity in data/Database.yaml (for example kg for CH4 and L for H2O). max_age is a calendar duration. BESS and interconnections may be used without defining Commodities_Production.

12 Outputs and reproducibility

12.1 output block

```
output:
  level: analysis
  event_threshold: 1.0e-09
  emissions: false
```

Field	Default / options	Meaning
level	analysis; comparison analysis detailed	Controls how much operational/time-series information is retained.
event_threshold	1e-9; >=0	Numerical zero used in event detection.
emissions	false	Calculate emissions. If true, every configured source must have emission_factor_co2.

12.2 Output levels

Level	Use
comparison	Compact scenario indicators; best for many-scenario comparison.
analysis	Recommended research default; adds temporal, event, storage, commodity and nuclear summaries.
detailed	Retains full time histories; use selectively because hourly Monte Carlo output can become large.

12.3 Main result files and interpretation

- Results.xlsx is the main human-readable workbook at every output level. Its available sheets include System, Generation, BESS, Commodities, Emissions, Ramps, Nuclear, Nuclear_Fuel, Convergence and Metadata, depending on the configured technologies.
- Tallies.xlsx is added at analysis and detailed levels and contains the more extensive temporal/event tables, including daily/monthly profiles, surplus and additional-generation-requirement episodes, storage/commodity operation and nuclear refueling/fuel statistics when applicable.
- Samples.parquet is the preferred compact per-simulation machine-readable table; if Parquet support is unavailable, LEAF-EB writes Samples.csv.gz instead.
- Diagnostics.xlsx is always written for the electricity-balance run and reports runtime diagnostics such as capacity clipping. Review it before interpreting a production ensemble.
- At detailed level, the Full/ directory is retained with full time-history products. At lower levels those detailed temporary histories are removed after consolidation.

Balance interpretation. Positive pre-flexibility balance is surplus. Negative pre-flexibility balance is the gross additional generation requirement. After the configured flexibility sequence, any remaining negative balance is the remaining additional generation requirement; its largest hourly magnitude is the peak remaining additional generation requirement.

Results.xlsx currently exposes these concepts with the metric names Initial_Surplus_Energy, Initial_Positive_Residual_Load_Energy, Remaining_Positive_Residual_Load_Energy and Peak_Remaining_Residual_Load. The latter three should be interpreted as gross/remaining/peak additional generation requirements, respectively.

12.4 Resolved_Config.yaml

LEAF-EB writes the effective configuration after defaults, technology-template merges, unit conversion and automatically resolved settings. Archive this file with published results. It also records provenance hashes where available.

12.5 Minimum reproducibility record

- Exact YAML input.

- Exact historical dataset or a stable data version/source.
- Resolved_Config.yaml.
- LEAF-EB release or Git commit.
- Monte Carlo seed, source list and simulation count.
- preserve_annual_targets and technology_uncertainty.
- Any technology database modifications.

12.6 Optional LEAF-to-ANICCA interface

For nuclear cases at analysis or detailed output level, deterministic simulation 0 writes ANICCA_Input.xlsx with one worksheet per nuclear fleet and the six-column structure Year | Installed_Power | Burnup | Load_Factor | Reactors_In | Reactors_Out. For burnup-driven fleets, Burnup is the configured target_burnup. For calendar-driven fleets, the automatic workbook leaves Burnup blank because a single annual value is not the same quantity as physical batch-discharge burnup.

When an annual-equivalent calendar burnup is explicitly needed for ANICCA, use the post-processing converter:

```
python scripts/leaf_to_anicca.py <scenario_output_dir> <input.yml> --statistic deterministic
```

The converter also accepts mean, p025, p50 and p975 for Monte Carlo annual series. Calendar-driven annual-equivalent burnup is an interface quantity; physical batch discharge burnup remains in LEAF-EB nuclear fuel outputs.

13 Public examples

Example	Adds	Use as starting point when...
Example_01_basic	Deterministic daily balance	You need the smallest complete scenario.
Example_02_monte_carlo	Temporal Monte Carlo	You need stochastic demand/generation series.
Example_03_bess	Hourly BESS	You need short-duration electrical storage.
Example_04_commodities	CH4 production/storage/reconversion	You need Power-to-X or long-duration commodity storage.
Example_05_nuclear_constant	Constant-output SMR + calendar refueling	You need explicit nuclear capacity and fixed-calendar outages.
Example_06_nuclear_load_following	Load-following SMR + burnup refueling	You need operation-dependent fuel exposure/refueling.

The examples are designed to be copied. Change `scenario.subfolder` first so a trial run cannot overwrite an existing example output.

Appendix A Complete YAML field reference

This appendix lists the canonical user-facing fields supported by the current schema. Fields generated internally, such as `hourly_patterns` inserted during normalization, are not intended to be written manually unless explicitly documented.

A.1 Top-level blocks

Path	Required	Accepted value
<code>scenario</code>	Yes	Configuration block
<code>historical_data</code>	Yes	Configuration block
<code>projection</code>	Yes	Configuration block
<code>simulation</code>	Yes	Configuration block
<code>demand</code>	Yes	Configuration block
<code>sources</code>	Yes	non-empty Configuration block
<code>commodities_input</code>	No	Configuration block
<code>output</code>	No	Configuration block

A.2 scenario / historical_data / projection

Path	Default / options	Rule
<code>scenario.folder</code>	required text	Results parent directory.
<code>scenario.subfolder</code>	required text	Scenario result directory.
<code>scenario.start_date</code>	required date	Start.
<code>scenario.end_date</code>	required date	Must not precede start.
<code>historical_data.file</code>	required path	CSV/XLSX.
<code>historical_data.date_column</code>	Date	Timestamp column.
<code>historical_data.processing_resolution</code>	daily hourly	Cannot request hourly processing from a daily dataset.
<code>historical_data.unit</code>	kWh MWh GWh TWh	Common energy unit for all historical energy columns.
<code>projection.resolution</code>	daily hourly	Hourly requires hourly processing.
<code>projection.pattern_file</code>	Pattern.xlsx	Normally leave default.
<code>projection.save_plots</code>	false	Strict YAML boolean.
<code>projection.variability.enabled</code>	false	Must be true for MC simulations.
<code>projection.variability.resolution</code>	projection resolution	Must match.
<code>projection.variability.mode</code>	global or M/monthly	Fallback-parameter grouping.
<code>projection.variability.plot_residuals</code>	false	Strict boolean.

A.3 simulation and Monte Carlo

Path	Default / options	Rule
<code>simulation.resolution</code>	daily hourly	Daily cannot consume hourly projection.
<code>simulation.capacity_tolerance</code>	0 %	0–100 %.
<code>simulation.console_output</code>	standard	quiet standard detailed.
<code>simulation.hourly_profile_file</code>	none	Needed for daily historical -> hourly simulation.

Path	Default / options	Rule
simulation.initial_condition s.bess_state_of_charge		0.0 0–1.
simulation.initial_condition s.commodity_inventory.<name>	0 if omitted	Non-negative.
simulation.monte_carlo.simulations		0 Non-negative integer.
simulation.monte_carlo.workers		1 Positive integer.
simulation.monte_carlo.seed		12345 Non-negative integer.
simulation.monte_carlo.confidence_level		0.95 Between 0 and 1.
simulation.monte_carlo.sources	required when simulations>0	Explicit list; no duplicates.
simulation.monte_carlo.preserve_annual_targets	false	Boolean.
simulation.monte_carlo.technology_uncertainty	false	Boolean.
simulation.monte_carlo.resume	false	Boolean.
simulation.monte_carlo.temporary_output_format	parquet	parquet pickle pkl.
simulation.monte_carlo.keep_analysis_temp	false	Boolean.
simulation.monte_carlo.save_perturbed_forecasts.first_monte_carlo	true	Boolean when mapping supplied.
simulation.monte_carlo.save_perturbed_forecasts.all_monte_carlo	false	Boolean when mapping supplied.

A.4 demand

Path	Required / options	Rule
demand.target_production	required	Energy per period; >0.
demand.model	linear exponential	custom is not used for demand.
demand.values	required	At least two non-negative dated factors.
demand.balance	default 1.0	>0.
demand.anchor.method	auto	auto mean median last max.
demand.anchor.window	1 when manual method used	Positive integer observations.

A.5 sources

Field	Allowed / requirement
share	Non-negative; required unless capacity_additions is defined.
capacity_factor	(0,1]; required when capacity_additions is absent.
capacity_tolerance	0–100 %; optional override.
model	linear exponential custom.
values	At least two dated non-negative factors for linear/exponential.
anchor	Optional method/window mapping.
limit	Optional non-negative production ceiling.

Field	Allowed / requirement
emission_factor_co2	Non-negative; explicit units supported; mandatory for all sources when output.emissions=true.
hourly_operation	profile_preserving dispatchable must_run load_following.
dispatch_priority	Positive integer where required; unique within operation class.
must_run	Mapping; power_fraction required for must_run.
load_following	Mapping with energy/control/power limits; extra fields for constrained_hourly.
unit_capacity	Positive power; required when refueling is defined.
refueling	Mapping. Exactly one calendar operating_cycle or burnup target basis; installed capacity must remain an integer multiple of unit_capacity. schedule defaults to auto; staggered enforces non-overlap when feasible.
fuel_cycle	Thermal power/core heavy-metal mass plus optional target_burnup/reference metadata. target_burnup is a full-residence/equilibrium target used to derive EFPD timing.
initial_capacity	Non-negative power for explicit-capacity custom source.
capacity_additions	Non-empty date -> non-zero power change mapping; requires model: custom.
reference_generation	capacity_factor mapping required for explicit capacity without refueling.
custom_mode	add replace; defaults to add for custom source.
custom_data	Non-empty date -> value mapping; mutually exclusive with capacity_additions.
replaces	Required list of replacement groups for custom_mode: replace.
technology_template	Mapping with file and name; scenario fields override template fields.

A.6 BESS and interconnections

Path	Options / rule
BESS.<name>.model	constant step linear.
BESS.<name>.value	Required for constant.
BESS.<name>.values	Dated power values for step/linear.
BESS.<name>.duration	Required; >0.
BESS.<name>.efficiency	Required; (0,1].
BESS.<name>.lifetime.calendar_years	Optional; >0.
BESS.<name>.lifetime.cycle_life_efc	Optional; >0.
Interconnections.<name>.mode	MW fraction; required.
Interconnections.<name>.import	Required mapping.
Interconnections.<name>.export	Required mapping.
import/export.model	constant step linear.
import/export.value	Required for constant.
import/export.values	Required for step/linear.
import/export.availability	Required; scalar 0–1 or complete month 1–12 mapping.

A.7 Commodities and dispatch

Path	Options / rule
commodities_input.run_commodities	Boolean. true runs the electricity-balance/flexibility engine; false (or an omitted commodities_input block) leaves only the constructed scenario series and skips BESS, interconnections and commodity dispatch.
commodities_input.database_path	Defaults to data/Database.yaml when commodities block is present.
Commodities_Production.<commodity>.share	Non-negative; production shares sum to 1.
Commodities_Production.<commodity>.technology	Must exist for commodity in Database.yaml.

Path	Options / rule
Commodities_Production.<commodity>.inputs	Must exactly match required upstream inputs in Database.yaml.
Commodity_Storage.<commodity>.max_age	Required; >0.
Commodity_Storage.<commodity>.max_storage_quantity	Required; >=0, in the commodity quantity unit defined by Database.yaml. Use this canonical field in new inputs.
Fuel_to_Electricity.<commodity>.share	Non-negative.
Fuel_to_Electricity.<commodity>.technology	Non-empty explicit technology.
Dispatch_Order.surplus	List using BESS, Interconnections, Commodities_Production; production last if present.
Dispatch_Order.deficit	List using BESS, Interconnections, Fuel_to_Electricity; reconversion last if present.

A.8 output

Path	Default / options	Rule
output.level	analysis	comparison analysis detailed.
output.event_threshold	1e-9	>=0.
output.emissions	false	Boolean; if true, all sources need emission_factor_co2.

Appendix B Accepted units

Quantity	Examples / accepted forms
Historical energy unit	kWh, MWh, GWh, TWh — one common unit per historical dataset.
Energy	500 MWh, 2 GWh, 0.2 TWh.
Energy per period	550 GWh/day, 200 TWh/year; hour/day/year denominators and common aliases.
Power	kW, MW, GW; MWe and MWth spellings are recognized where the field already defines physical meaning.
Duration	h/hour and day for elapsed durations.
Calendar cycle	month or year.
Heavy-metal mass	kgHM or tHM. Ambiguous ton/tons are rejected.
Burnup	MWd/tHM, GWd/tHM, MWd/kgHM, GWd/kgHM.
Emission factor	gCO2e/kWh, kgCO2e/MWh and equivalent supported combinations.
Percentage	%, percent, pct.
Fractional rate	/h, /hour, /day and equivalent 1/... forms.
Commodity quantity	Uses Commodity_Units in data/Database.yaml; e.g. CH4/H2/NH3/N2/CO2 in kg and H2O in L. Storage capacity and initial inventory use this unit.

Dimensional rule. New public dimensional fields should carry explicit units. For example, write "300 MW" rather than 300 for capacity_additions.

Appendix C Common errors and validation rules

Symptom / message	Cause	Correction
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Symptom / message	Cause	Correction
Missing required input sections	One of scenario, historical_data, projection, simulation, demand or sources is absent.	Start from Example_01_basic and retain all required blocks.
Historical data is missing required columns	A standard source name is not present in the dataset.	Rename the historical column or the source consistently; custom sources do not require a historical column.
Hourly processing requires hourly historical data	processing_resolution=hourly but the file is daily.	Use daily processing or provide hourly history.
Hourly simulation ... requires simulation.hourly_profile_file	Daily historical data are being expanded to hourly operation without a profile.	Provide a valid 8760-row hourly profile.
values must contain at least two dated values	linear/exponential model has too few milestones.	Provide at least two dates.
capacity_factor is required	Standard source has no explicit capacity schedule and no capacity factor.	Add a physically justified capacity_factor.
capacity_additions currently requires model: custom	Explicit capacity schedule used with linear/exponential model.	Set model: custom.
reference_generation must be omitted when refueling is enabled	Both reference_generation and refueling were supplied.	Remove reference_generation; refueling availability defines the reference.
must_run.power_fraction is required	must_run selected without its physical output fraction.	Add must_run: {power_fraction: ...}.
load_following requires ...	Required load-following fields are missing.	Provide energy_policy, control_mode, minimum and maximum fractions; add constrained fields when needed.
must define exactly one refuelling basis	Both operating_cycle and target_burnup are present, or neither is present.	Keep exactly one basis.
Monte Carlo requires at least one eligible stochastic energy series	simulations>0 but sources list is empty.	Explicitly list Demand and/or configured generation sources.
BESS duration/efficiency is required	Storage physics is incomplete.	Always define both duration and efficiency.
Dispatch_Order ... final stage	Commodity production/reconversion is not last.	Move Commodities_Production/Fuel_to_Electricity to the end of the relevant list.
output.emissions is true, but emission_factor_co2 is missing	At least one source lacks an emissions factor.	Define explicit factors for every source or set emissions:false.
Installed capacity ... must be an integer multiple of unit_capacity	A refueling fleet implies a fractional equivalent reactor unit.	Use capacity additions/retirements that keep total fleet MW at exact multiples of unit_capacity.
No electricity-balance results are produced	commodities_input is absent or run_commodities is false.	Add commodities_input: {run_commodities: true}. No BESS or commodity pathway is required for a balance-only case.
Custom replacement is unallocated	The ordered replacement groups do not contain enough annual generation to absorb the custom source reference energy.	Review the replaces groups or deployment. LEAF-EB reports the unmatched part rather than forcing negative generation.

Appendix D Complete starting templates

D.1 Minimal deterministic scenario

```

scenario:
  folder: Outputs
  subfolder: MyScenario
  start_date: '2030-01-01'
  end_date: '2050-12-31'

historical_data:
  file: Inputs/data/my_history.csv
  date_column: Date
  processing_resolution: daily
  unit: MWh

projection:
  resolution: daily
  variability:
    enabled: false
  save_plots: false

simulation:
  resolution: daily
  capacity_tolerance: "0 %"
  console_output: standard

demand:
  target_production: "200 TWh/year"
  model: linear
  balance: 1.0
  values:
    '2030-01-01': 0.70
    '2050-01-01': 1.00

sources:
  Wind:
    share: 0.40
    capacity_factor: 0.38
    model: linear
    values:
      '2030-01-01': 0.70
      '2050-01-01': 1.00
  Solar:
    share: 0.35
    capacity_factor: 0.22
    model: linear
    values:
      '2030-01-01': 0.70
      '2050-01-01': 1.00
  Firm:
    share: 0.25
    capacity_factor: 0.70
    model: linear
    values:
      '2030-01-01': 0.70
      '2050-01-01': 1.00

commodities_input:
  run_commodities: true

output:
  level: comparison
  event_threshold: 1.0e-09
  emissions: false

```

D.2 Add Monte Carlo

```

projection:
  variability:
    enabled: true
    resolution: daily
    mode: M
  plot_residuals: false

simulation:
  monte_carlo:
    simulations: 200
    workers: 4
    seed: 12345

```

```
confidence_level: 0.95
preserve_annual_targets: false
technology_uncertainty: false
resume: true
sources: [Demand, Wind, Solar, Firm]
```

D.3 Add an hourly BESS

```
simulation:
  resolution: hourly
  initial_conditions:
    bess_state_of_charge: 0.0

commodities_input:
  run_commodities: true
  BESS:
    Total:
      model: constant
      value: "1000 MW"
      duration: "4 h"
      efficiency: 0.90
  Dispatch_Order:
    surplus: [BESS]
    deficit: [BESS]
```

End of manual.